

Msc Thesis 1 Page Summary

Title: Bayesian reinforcement-learning models of cognitive flexibility and the effects of over-training

Author: Salaar Mir — MSc Computational Neuroscience, Cognition & AI (Univ. of Nottingham)

Research question

- How does extensive training (over-training) affect cognitive flexibility in agents modelled with Bayesian reinforcement learning? I tested whether Bayesian RL agents replicate the “over-training enhancement” (ORE) reported in some animal studies, or whether over-training produces more habitual, less flexible behaviour.

Method (concise)

- Simulations implemented in Python (NumPy/Pandas/Matplotlib); code available on GitHub [link in thesis].
- Task: two-choice reversal learning (choose left/right). Rules reversed after 10 consecutive correct trials; over-trained agents received +100 trials after criterion. Four rules (three reversals) per run. 50 iterations per condition were averaged.
- Model: Bayesian RL with Beta priors (uniform init). Action selection: Thompson Sampling (default) and Bayes-UCB compared. Key parameters: decay factor $\gamma \in \{0.5, 0.7, 0.9\}$, noise $\eta \in \{0.1, 0.2, 0.3\}$. Metrics: trials-to-learn per rule, MAP action probabilities, cumulative rewards/choices. Statistical test: three-way ANOVA (Decay $\times$ Noise $\times$ Over-training).

Key quantitative results

- Both normally trained and over-trained agents learned initial rule comparably; over-trained agents required substantially more trials to adapt after reversals (no ORE replicated).
- Example extremes: fastest normally trained agent ($\gamma=0.9, \eta=0.1$) learned all rules in 110 trials; fastest over-trained agent ($\gamma=0.9, \eta=0.1$) learned all rules in 603 trials (excluding extra training trials).

- ANOVA: Noise effect significant ($F=3.363$, $p=0.042$); Over-training significant ($F=16.654$, $p=0.000149$); Decay factor not significant ($p=0.561$). No significant interactions reported.
- Bayes-UCB vs Thompson: under tested settings ($\gamma=0.9$, $\eta=0.2$) produced similar cumulative choice distributions — algorithm choice had minimal effect in this environment MSc_Thesis.pdf.

Interpretation & contribution

- Over-training in this Bayesian RL formulation produced more peaked belief distributions (Beta priors), leading to habitual, slower re-learning after reversals — consistent with habit formation rather than ORE.
- The study shows Bayesian RL can capture habit-like effects, and it highlights a mismatch with some empirical ORE findings, suggesting that meta-learning / hierarchical inference may be required to capture the empirical improvement over repeated reversals.
- Contribution: a systematic simulation study mapping parameter regimes for over-training effects; clear, reproducible codebase; and a set of concrete follow-ups for PhD research.

Limitations

- Simple binary task; no stochastic rewards (deterministic reversals used).
- No explicit meta-learning/hierarchical model implemented (agents do not infer task structure / switch probability).
- Limited baseline comparisons in current version (recommend Q-learning/softmax baselines and sensitivity to prior choices).

Clear next steps for PhD work

- Implement hierarchical/meta-learning agents (e.g., hierarchical Bayesian, MAML, recurrent controllers) to test whether structure-learning produces ORE.
- Introduce stochastic rewards and richer tasks, compare to Q-learning / model-based baselines, and perform broader sensitivity analyses (priors, seeds, iteration counts).
- Apply to perception/robotics tasks (sim-to-real) where adaptive learning across rule changes matters for autonomy.

